Priority Based Selection Hyperparameter Analysis

Fleet maintenance optimisation

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Main finding

The recommended settings for the as-loaded forecast are half steps with first level selection and max_iters equal to 400. They achieve worst-day availability of 18 aircraft and smoothness of 1.391860, a 1.56 percent reduction in standard deviation from the default. Mean availability remains 21.713010 aircraft.

Worst-day availability of 18 is globally maximal under the first D check rules for this forecast. Six aircraft must be in D maintenance together on six dates in May 2033. Tuning therefore improves the distribution of availability around an already optimal worst day. The best smoothness found is an empirical result; the study does not establish a global smoothness optimum.

Schedule	Worst day	Std deviation	Mean
As loaded	15	1.949767	21.713010
Default	18	1.413971	21.713010
Previous tuned baseline	18	1.391860	21.713010
Best in extended study	18	1.391860	21.713010

The analysis combines 74 historical runs with 34 fresh runs, covering 87 distinct parameter sets and four start schedules. It examines iteration budgets, shift magnitudes, ladder structure, ladder order, selection policy, interactions, local refinements, runtime variability, and sensitivity to legal changes in the start schedule.

Scope: savn_priority_based_selection.py on heavy_maintenance_forecast.xlsx, MxEvents. The simulated annealing and early D variants are outside this comparison.

Algorithm and hyperparameters

The baseline is a deterministic hill-climb using a sequence of neighbourhoods. For each neighbourhood, checks are ordered by the lowest availability during their active maintenance interval. The search scans priority levels from worst to best and stops at the first level containing an improving move. It then accepts the best move within that level and starts the next iteration at the first neighbourhood again.

A proposal moves one check by an offset and rebuilds the aircraft's later checks. Rebuilds enforce the maintenance sequence and recalculate due dates and legal start windows. Positive offsets may move a check later within its legal window; no check may start after its current due date. A larger improving shift can cross over schedules that are worse, but the algorithm never accepts a worsening endpoint.

Setting	Default	Range analysed
Iteration cap	400	5 to 800 plus full trace checkpoints
Shift magnitude in days	28 14 7 3	1 to 70
Neighbourhood structure	Four pairs	Pairs half offsets unions asymmetric sets
Neighbourhood order	Large to small	Reversed interleaved and refined ladders
Selection policy	First priority level	First level all levels first improving

What is and is not tunable

Only max_iters is exposed by savn_optimise. shift_sets is a local constant in that function, and first-priority-level selection is built into generate_neighbors. This study varies these algorithm settings in a separate driver. Temperature, cooling rate, tabu tenure, and a search random seed do not apply to this deterministic method.

Objective and fixed inputs

Scores are compared lexicographically as (minimum availability, negative standard deviation, mean availability). A gain of one aircraft on the worst day outranks any smoothness change. Standard deviation uses the population definition implemented by numpy.std. No weighted scalar objective is introduced.

Maintenance rules are fixed domain inputs: A 14 day duration 365 day interval 42 day early allowance; B 28 730 70; C 84 1460 70; D 126 2920 70. Changing these rules would change the scheduling problem rather than tune the optimiser.

Dataset and experimental design

Property	Value
Aircraft and checks	24 aircraft and 278 checks
Check counts	A 144 B 72 C 38 D 24
Observation dates	2026-01-21 to 2038-04-28
Forecast horizon	Day indices 0 to 4480 inclusive and 4481 observations
Total scheduled maintenance	10248 aircraft days
Starting score	15 and 1.949767 and 21.713010

The historical evidence contains 74 run records and 72 distinct parameter sets. Duplicate baseline settings and different caps are retained for auditability. The historical file dates from 17 September 2026; the extension was executed on 13 October 2026. Fresh default and half-step runs reproduce the historical scores to numerical precision. The historical study did not store input hashes, so matching input statistics and reproduced endpoints establish consistency, rather than a verified historical byte identity.

All comparisons use the same horizon fixed from the as-loaded schedule. The extension uses an 800-iteration safety cap to distinguish convergence from a budget limit. The final recommendation uses the measured convergence requirement. Historical caps, including 400, remain labelled with their own values in the full catalogue.

Nine serial finalist repetitions measure elapsed time for the default, half steps, and wide-first ladder. Any new leading configuration receives further repetitions and start-schedule checks. Repeating a deterministic run measures runtime variability, not search quality variability. Three separately generated start schedules use fixed seeds 41, 42, and 43 solely to perturb legal starts.

Validation

The production default and study driver return the same complete schedule, score, and trace length. Every fresh endpoint passes complete maintenance-rule replay and a recomputation of daily availability. A separate boolean occupancy calculation verifies availability without the delta-array method. Every accepted score strictly improves, check counts and durations are conserved, and all 74 historical endpoints are marked rule-compliant. The nine existing baseline tests also pass in the analysis runtime.

Runtime: Python 3.12.14 with NumPy 2.3.5 on Windows-11-10.0.26200-SP0. GUI plotting imports are omitted when loading the unchanged production scheduling functions into the bundled analysis runtime.

Iteration budget and convergence

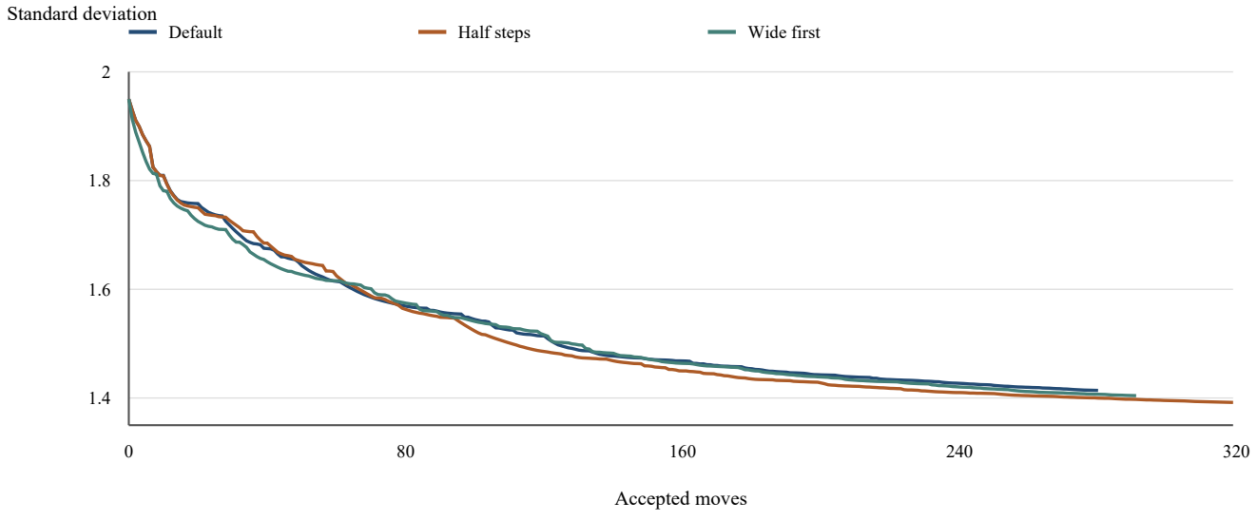


Figure 1 Smoothness along the deterministic search path. Each point is an accepted move; the horizontal axis is not elapsed time.

Cap	Default min	Default std	Half min	Half std
5	16	1.872941	16	1.872941
10	17	1.809552	17	1.808565
20	17	1.757626	17	1.749992
40	17	1.674928	18	1.684892
80	17	1.568984	18	1.562999
160	17	1.468172	18	1.449818
200	18	1.442565	18	1.427636
280	18	1.413971	18	1.399854
319	18	1.413971	18	1.391860
400	18	1.413971	18	1.391860
800	18	1.413971	18	1.391860

Checkpoint results are exact prefixes of the measured trajectories, rather than extra full runs. The default converges after 280 accepted moves and half steps after 319. An additional full pass detects exhaustion. Increasing a cap beyond natural convergence cannot improve an endpoint because the stopping rule still terminates the search.

Default first reaches 18 after 166 accepted moves; Half steps first reaches 18 after 40 accepted moves; Wide first first reaches 18 after 233 accepted moves. Caps that stop before that point sacrifice the primary objective. A 400 cap provides headroom for the as-loaded finalists; start variants may need more.

Sensitivity to individual shift magnitudes

Offset in days	Worst day	Std deviation	Moves	Exhausted
1	17	1.765355	400	No
2	17	1.664235	400	No
3	17	1.590036	400	No
5	18	1.490649	400	No
7	18	1.470754	334	Yes
10	17	1.465586	254	Yes
14	18	1.474391	188	Yes
21	17	1.525571	128	Yes
28	17	1.554409	96	Yes
35	17	1.581875	89	Yes
42	17	1.607762	67	Yes
56	16	1.704382	28	Yes
70	15	1.937597	2	Yes

A single plus-or-minus seven-day neighbourhood is the strongest converged single-magnitude setting in this sweep. It reaches 18 available aircraft but ends at standard deviation 1.470754, worse than the default and tuned ladders. A fourteen-day neighbourhood also reaches 18 quickly, after 188 accepted moves, but leaves standard deviation 1.474391.

The one-, two-, and three-day searches hit the 400-move cap at worst-day availability 17. These results show inefficient progress within the tested budget; they do not establish that the small-step neighbourhoods would converge at 17. The five-day run also hits its cap, even though it reaches 18.

Very wide single steps are poor on this workbook. The seventy-day setting accepts only two moves and converges at 15, while fifty-six days converges at 16. Wide jumps often lie outside legal windows or offer no improving endpoint. That explanation is consistent with the outcome but the historical study did not count rejected proposals, so it is not a measured rejection-rate result.

Implication for tuning

Retain multiple scales of movement. The strongest single neighbourhood does not replace a ladder, and larger offsets do not monotonically improve the result. Fine offsets are useful for later smoothness improvement when combined with earlier larger shifts.

Source: historical single_neighbourhood phase. A No in Exhausted means the endpoint remains budget-limited and should not be described as a local optimum.

Neighbourhood ladders and structure

Ladder	Worst day	Std deviation	Moves
Dense	18	1.396502	329
Geometric plus day	18	1.399695	346
Wide dense	18	1.402084	339
Plus wide	18	1.404311	291
Geometric	18	1.404470	333
Plus day	18	1.407168	309
Plus month	18	1.407327	319
Monthish	18	1.407644	262
Early window	18	1.409862	303
Fine	18	1.411760	341
Default	18	1.413971	280
Weekly	18	1.415076	316
Three level	18	1.416179	311
Two level 14 3	18	1.420113	315
Two level 28 7	18	1.436985	215
Coarse	17	1.524547	137

The dense ladder of plus-or-minus 28, 21, 14, 7, 3, and 1 days is the best ordinary paired ladder in the historical sweep. Its standard deviation is 1.396502. A geometric ladder of 32, 16, 8, 4, and 2 days improves from 1.404470 to 1.399695 when a one-day final neighbourhood is added.

The coarse 70, 42, and 28 day ladder converges at worst-day availability 17. Two-level shortcuts reach 18 but lose smoothness. These observations support a mixture of coarse movement and fine finishing for this forecast. The exact ranking is path-dependent: the algorithm takes the first neighbourhood with an improvement, so adding or reordering a neighbourhood changes later schedules.

Half offsets within each neighbourhood

The previous tuned baseline uses four neighbourhoods: (-28, -14, 14, 28), (-14, -7, 7, 14), (-7, -3, 3, 7), and (-3, -1, 1, 3). The best improving move is selected among all offsets within the first improving priority level. This produces a different trajectory from trying each offset pair in a separate neighbourhood.

A single union of the original offsets returns standard deviation 1.413656. Adding 21- and one-day offsets to that union reaches 1.406534. Both are worse than the half-step ladder. This is measured evidence for retaining the layered search structure on this input, rather than a general guarantee that layering always wins.

Order sensitivity and neighbourhood ablations

Order setting	Cap	Worst day	Std deviation	Exhausted
order small first default	400	17	1.590036	No
order small first weekly	400	17	1.765355	No
order interleaved	400	18	1.408437	Yes
order wide then day	400	17	1.488401	No
reverse halves	800	18	1.438537	Yes
reverse dense	800	17	1.652663	No

Historical small-first default and weekly ladders both stop at their 400 cap with worst-day availability 17. Their endpoints closely resemble the corresponding small single-step searches because the first small neighbourhood keeps providing improving moves. Fresh reverse-order experiments use a higher cap and therefore test whether the earlier result was merely insufficient budget.

Reversed half steps reaches 18 available aircraft and ends at standard deviation 1.438537 after 792 moves. The reversed dense ladder ends at 17 with standard deviation 1.652663 after 800 moves. It remains cap-limited, so its eventual endpoint is unresolved.

Removed extra offsets	Worst day	Std deviation	Change from half steps
Remove half offsets at level 1	18	1.408595	0.016735
Remove half offsets at level 2	18	1.395383	0.003523
Remove half offsets at level 3	18	1.396183	0.004322
Remove half offsets at level 4	18	1.400332	0.008472

Each ablation restores one half-step neighbourhood to its original plus-or-minus pair, leaving the other three unchanged. A positive change in standard deviation means worse smoothness at the same worst day. This comparison isolates each local change in settings, although the resulting search path can differ from the first move onward.

Ablations should be read together rather than added. The effect of removing offsets at one level depends on which other offsets remain available. The study measures these conditional effects and does not assign an independent contribution to each level.

The strongest ablation is remove half offsets at level 2, with standard deviation 1.395383. Its change relative to the full half-step ladder is +0.003523. This is a conditional comparison at the same worst-day availability, not evidence that the removed offsets are always beneficial or harmful.

The original small-first claims apply to a 400 cap. Fresh reversed ladders are labelled separately with an 800 cap to avoid conflating ordering and stopping budget.

Selection policy interactions

Neighbourhoods	Selection	Worst day	Std deviation	Moves
Default	first level	18	1.413971	280
Default	all levels	18	1.415233	286
Default	first improving	18	1.421370	314
Half steps	first level	18	1.391860	319
Half steps	all levels	18	1.403198	310
Half steps	first improving	18	1.408753	391
Dense pairs	first level	18	1.396502	329
Dense pairs	all levels	18	1.401448	326
Dense pairs	first improving	18	1.404629	344

First level returns all improving candidates in the first priority level that contains any, then takes the best. All levels takes the best improving candidate across every priority level for the active neighbourhood. First improving stops at the first improving candidate in priority level, start date, aircraft name, check index, and offset order. All three still require a strictly better lexicographic score.

The interaction table tests each policy on three different neighbourhood structures. This avoids assuming that a selection rule measured on the default transfers unchanged to the tuned ladder. Ties in candidate scores retain the first candidate in the deterministic iteration order.

The best endpoint in this interaction comparison uses half steps with first level selection and standard deviation 1.391860. The report recommends settings based on the complete extended comparison, including local refinements, rather than treating the selection comparison as a separate optimisation target.

Priority sorting directs proposals towards checks that cover low-availability days, but it is a search heuristic. Even all-level selection preserves worst-day availability through the score rule. Any smoothness differences arise from different accepted paths, not from one policy allowing worse worst days.

Fresh results use the same 800 safety cap. Historical default all-level selection ended at 1.415233; half-step all-level selection ended at 1.403198. Reproduced endpoints provide an additional consistency check.

Local refinement and the recommended settings

Setting	Worst day	Std deviation	Moves	Exhausted
Half steps	18	1.391860	319	Yes
Half steps plus two days	18	1.391860	319	Yes
Half steps plus 42 days	18	1.405105	287	Yes
Half steps plus 21 days	18	1.396183	321	Yes

The local refinement adds a two-day final pair, a forty-two-day first pair, or twenty-one-day offsets within the largest half-step neighbourhood. These are targeted nearby alternatives to the previous winner. Each changes one structural feature while preserving the maintenance rules and acceptance rule.

Recommended configuration for the as loaded forecast

Half steps is the strongest lexicographic endpoint in the fresh as-loaded comparison. It accepts 319 moves, converges naturally, and finishes at worst-day availability 18 with standard deviation 1.391860. Relative to the default, the absolute reduction in standard deviation is 0.022111 aircraft, or 1.56 percent.

Neighbourhood 1 -28, -14, +14, +28 days

Neighbourhood 2 -14, -7, +7, +14 days

Neighbourhood 3 -7, -3, +3, +7 days

Neighbourhood 4 -3, -1, +1, +3 days

Selection first level

Use max_iters equal to 400. The largest observed move count for this configuration across the as-loaded and three perturbed starts is 319. Stop naturally when a full pass finds no improving candidate and report whether the cap was reached. The fresh study used an 800 cap to check that the recommendation was not a budget-limited endpoint.

The recommendation applies to this forecast and these rules. The controlled start checks on the next page assess sensitivity, while wider generalisation still requires forecasts from other fleet sizes and maintenance distributions.

Sensitivity to the start schedule

Variant	Changed aircraft	Initial worst day	Initial std
As loaded	0	15	1.949767
41	24	15	1.947247
42	24	15	1.953883
43	24	15	1.932522

For each fixed seed, the analysis attempts one nonzero legal shift of up to fourteen days on a later check of every aircraft, then rebuilds that aircraft. The first due clocks, check sequence, check count, durations, fleet, and forecast horizon remain fixed. Later due dates respond to rebuilds under the same maintenance rules. These variants test local start sensitivity; they are not independent datasets.

Seed	Configuration	Worst day	Std deviation	Moves	Exhausted
41	Default	18	1.422469	280	Yes
41	Half steps	18	1.405264	299	Yes
41	Wide first	18	1.405740	294	Yes
42	Default	18	1.415864	305	Yes
42	Half steps	18	1.403357	318	Yes
42	Wide first	18	1.409704	292	Yes
43	Default	18	1.412076	272	Yes
43	Half steps	18	1.400651	307	Yes
43	Wide first	18	1.413182	272	Yes

Default has median smoothness 1.415864, ranging from 1.412076 to 1.422469 across the three perturbed starts. Worst-day availability ranges from 18 to 18.

Half steps has median smoothness 1.403357, ranging from 1.400651 to 1.405264 across the three perturbed starts. Worst-day availability ranges from 18 to 18.

Wide first has median smoothness 1.409704, ranging from 1.405740 to 1.413182 across the three perturbed starts. Worst-day availability ranges from 18 to 18.

The strongest measured configuration within each start variant is Half steps on 3 of three variants. A setting that wins on the as-loaded forecast need not win on every nearby start.

Compare configurations within the same variant. A range across three nearby starts does not support a statistical confidence interval or a population robustness claim.

Runtime and practical tradeoffs

Configuration	Runs	Median seconds	Range seconds	Std deviation	Moves
Default	3	31.74	25.13 to 49.08	1.413971	280
Half steps	3	57.71	51.35 to 61.84	1.391860	319
Wide first	3	44.31	31.87 to 57.26	1.404311	291

Every repeated finalist returns an identical full schedule. Elapsed times measure the quiet study driver, including copying and trajectory statistics but excluding the final independent audits, file writes, plotting, and exports. The production default equivalence run is a validation step outside these timing samples. No experiment workers ran concurrently.

Three timings give a useful local median and range, but host load and power management remain uncontrolled. Historical timing values span very different repetitions; for example, a geometric plus one-day refinement took 987 seconds while an earlier run took 95 seconds. The analysis therefore does not compare fresh medians with individual historical timings or rank all 74 historical settings by speed.

Half steps requires 1.82 times the default median runtime on this machine. Its standard deviation changes by 1.56 percent at the same worst-day availability of 18.

Wide first requires 1.40 times the default median runtime on this machine. Its standard deviation changes by 0.68 percent at the same worst-day availability of 18.

Choosing an operating point

Use the best measured configuration for offline planning when its extra runtime is acceptable. Wide first is a modest change to the original ladder and offers a smaller smoothness gain. The default remains the frozen baseline for method comparisons. Select runtime and quality tradeoffs using reproduced timings rather than a historical outlier.

Effects on daily availability and displacement

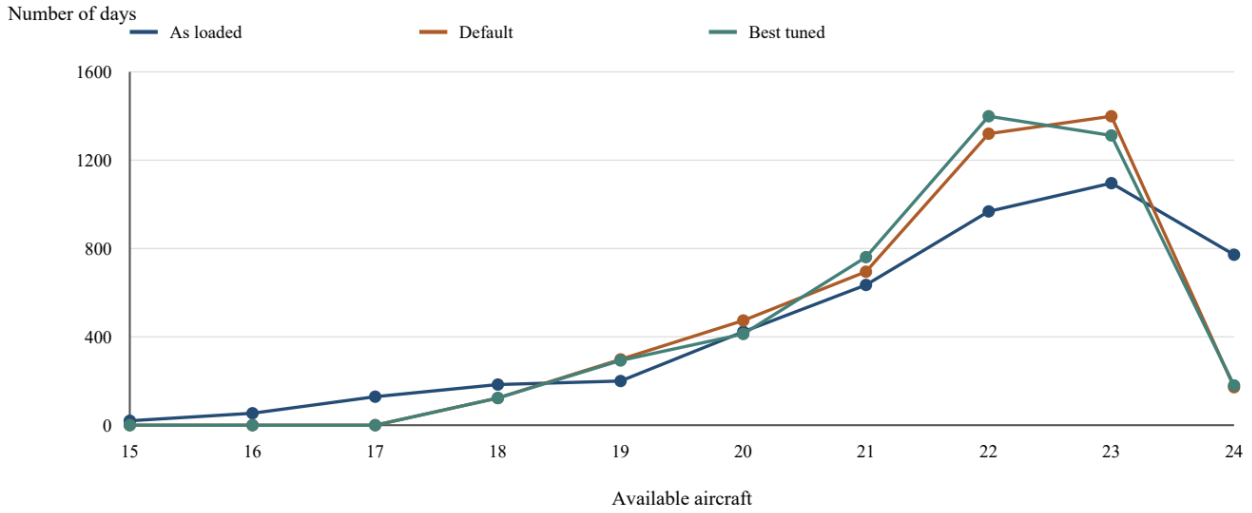


Figure 2 Distribution of daily availability. Connected markers guide comparison between integer availability levels.

Schedule	Days below 19	Days at worst value	Std deviation
As loaded	387	20	1.949767
Default	123	123	1.413971
Best tuned	123	123	1.391860

Worst-day frequency and standard deviation answer different questions. Minimising standard deviation at a fixed worst day can redistribute days among several availability levels. The score does not explicitly minimise the number of days below 19, so that count must be reviewed alongside smoothness rather than inferred from it.

In this comparison, the default has 123 days below 19 and the best tuned schedule has 123. The smoothness improvement therefore comes from redistributing availability on other days while retaining the same count of low-availability days.

Check	Count	Moved	Earliest shift	Latest shift	Median absolute shift
A	144	92	-72	69	7.0
B	72	50	-58	45	10.5
C	38	33	-120	49	14.0
D	24	23	-62	69	19.0

Start shifts are days relative to the as-loaded forecast and include rebuild effects. Positive values remain inside recalculated legal windows. There is no displacement limit in this baseline.

Availability limits and operational scope

The mean is fixed by conserved total downtime. Over 4481 days, the 278 checks require 10248 aircraft days of maintenance, so mean availability is 24 minus 10248 divided by 4481, or 21.713010489. Changing only legal starts cannot improve the mean when all checks stay inside the comparison horizon. The third score component therefore does not discriminate among these runs.

Why worst day availability cannot exceed 18

Each aircraft's first D due date is its first A due date plus 2555 days. A, B, and C checks do not reset the D due clock. The first D duration is 126 days and its early allowance is 70 days. A D check with due day d can start only between d minus 70 and d. Every legal placement occupies the forced interval from d to d plus 56, with the end excluded.

Aircraft	D due date	Last forced day
Aircraft 11	01 May 2033	25 Jun 2033
Aircraft 14	05 May 2033	29 Jun 2033
Aircraft 15	31 Mar 2033	25 May 2033
Aircraft 19	20 May 2033	14 Jul 2033
Aircraft 24	16 Apr 2033	10 Jun 2033
Blue-8	02 Apr 2033	27 May 2033

These six forced intervals overlap on 20 to 25 May 2033. At least six aircraft must be unavailable on each of these dates, so at most 18 can be available. Since the tuned schedules achieve a worst day of 18 across the whole horizon, the primary objective is proven optimal for this forecast and the unchanged first D rules.

This bound does not prove the best possible smoothness at minimum 18. A different neighbourhood or a search that accepts intermediate worsening schedules could still improve standard deviation. Reaching 19 would require changing a problem assumption, such as the D early allowance, first due clocks, or check duration, and must be treated as a separate scenario.

Operational constraints

The baseline validates maintenance rules, legal starts, non-overlap, and the horizon. It does not load operational_constraints.json and does not enforce busy periods, dependency groups, or displacement limits. A tuned schedule must be re-evaluated under any required operational constraints before it is used as an operational plan. Hyperparameter rankings under added constraints may differ.

Implementation and reproducibility

Applying the recommendation

The production module and source workbook remain unchanged. To apply the recommended neighbourhoods in savn_optimise, replace the local shift_sets list with the neighbourhoods listed on the local refinement page. Preserve maintenance rebuilds and lexicographic acceptance. If the recommended policy is first level, no generate_neighbors change is needed; alternative policies require the corresponding study selection behaviour to be made explicit.

Expose shift_sets as a function argument for repeatable comparison rather than editing a local constant for every experiment. Keep the frozen default as the default argument for baseline comparisons. Record the chosen cap, the accepted-move count, and whether the search stopped naturally. An 800 cap provides room for the observed start-schedule sensitivity experiments without requiring 800 moves.

Reproducing the evidence

Run .scratch/priority-hyperparameter-analysis/run_analysis.py with the Python runtime listed in the source record. The script resumes completed trials only when source workbook and production hashes match. To force a clean rerun, use a fresh study folder or preserve and rename the existing fresh-results.json before running again. The original study also skips existing names and must not be mistaken for a fresh execution when rerun against its stored file.

fresh-results.json contains complete endpoint metrics, daily distributions, hashes, and accepted-move trajectories. Each fresh run also has a schedule JSON; start_41.json, start_42.json, and start_43.json preserve the sensitivity inputs. analysis-summary.json records combined counts and the selected configuration. The historical run table and fresh catalogue follow in the appendices.

Limits of the recommendation

This study covers one 24-aircraft forecast and three small legal perturbations of it. It does not test an independent fleet, a different horizon, larger data, changed maintenance rules, or operational constraints. The search is deterministic and the setting sweep is deliberately selected rather than exhaustive. The measured best is a tuned baseline for this problem, and future method comparisons should report comparable outcomes and budgets against it.

The start-variant medians describe sensitivity across three fixed inputs. Timing medians describe three repeated executions of one deterministic input. Neither supports statistical significance claims. Report a cap-limited endpoint as incomplete convergence and avoid selecting settings solely from smoothness when their worst-day availability is lower.

Historical run catalogue

All 74 historical records are retained, including repeated settings and cap refinements. Standard deviation is in aircraft. Exhausted indicates that a full neighbourhood pass found no improving move. Seconds are historical observations and should not be compared directly with the fresh timing medians.

ID	Run name	Cap	Min	Std	Moves	End	Seconds
H01	baseline default	400	18	1.413971	280	Local	14.6
H02	iters 5	5	16	1.872941	5	Cap	0.0
H03	iters 10	10	17	1.809552	10	Cap	0.0
H04	iters 20	20	17	1.757626	20	Cap	0.1
H05	iters 40	40	17	1.674928	40	Cap	0.3
H06	iters 80	80	17	1.568984	80	Cap	1.0
H07	iters 160	160	17	1.468172	160	Cap	4.4
H08	iters 400	400	18	1.413971	280	Local	14.9
H09	iters 800	800	18	1.413971	280	Local	15.0
H10	single pm1	400	17	1.765355	400	Cap	3.3
H11	single pm2	400	17	1.664235	400	Cap	4.2
H12	single pm3	400	17	1.590036	400	Cap	8.2
H13	single pm5	400	18	1.490649	400	Cap	13.6
H14	single pm7	400	18	1.470754	334	Local	23.3
H15	single pm10	400	17	1.465586	254	Local	5.3
H16	single pm14	400	18	1.474391	188	Local	3.8
H17	single pm21	400	17	1.525571	128	Local	2.4
H18	single pm28	400	17	1.554409	96	Local	1.5
H19	single pm35	400	17	1.581875	89	Local	1.3
H20	single pm42	400	17	1.607762	67	Local	0.9
H21	single pm56	400	16	1.704382	28	Local	0.3
H22	single pm70	400	15	1.937597	2	Local	0.0
H23	vns default	400	18	1.413971	280	Local	15.2
H24	vns plus day	400	18	1.407168	309	Local	20.3
H25	vns plus wide	400	18	1.404311	291	Local	19.5
H26	vns plus month	400	18	1.407327	319	Local	23.1
H27	vns early window	400	18	1.409862	303	Local	20.6
H28	vns weekly	400	18	1.415076	316	Local	19.6
H29	vns fine	400	18	1.411760	341	Local	14.8
H30	vns coarse	400	17	1.524547	137	Local	4.0
H31	vns geometric	400	18	1.404470	333	Local	22.4
H32	vns geometric plus day	400	18	1.399695	346	Local	94.9
H33	vns three level	400	18	1.416179	311	Local	58.2
H34	vns two level 28 7	400	18	1.436985	215	Local	25.8
H35	vns two level 14 3	400	18	1.420113	315	Local	37.7
H36	vns monthish	400	18	1.407644	262	Local	47.6
H37	vns dense	400	18	1.396502	329	Local	82.7
H38	vns wide dense	400	18	1.402084	339	Local	28.9

ID	Run name	Cap	Min	Std	Moves	End	Seconds
H39	order small first default	400	17	1.590036	400	Cap	5.7
H40	order small first weekly	400	17	1.765355	400	Cap	3.0
H41	order interleaved	400	18	1.408437	343	Local	13.6
H42	order wide then day	400	17	1.488401	400	Cap	24.3
H43	rich paired weeks	400	18	1.403516	261	Local	41.6
H44	rich one union	400	18	1.413656	254	Local	42.4
H45	rich dense union	400	18	1.406534	282	Local	66.5
H46	rich vns with halves	400	18	1.391860	319	Local	96.5
H47	asymmetric left heavy	400	18	1.419956	274	Local	34.5
H48	asymmetric right heavy	400	18	1.412866	290	Local	38.3
H49	select all levels default	400	18	1.415233	286	Local	69.8
H50	select first improving default	400	18	1.421370	314	Local	45.5
H51	refine rich vns with halves iters 80	80	18	1.562999	80	Cap	3.6
H52	refine rich vns with halves iters 200	200	18	1.427636	200	Cap	20.0
H53	refine rich vns with halves iters 800	800	18	1.391860	319	Local	34.9
H54	refine rich vns with halves all levels	400	18	1.403198	310	Local	49.7
H55	refine vns dense iters 80	80	17	1.568984	80	Cap	1.1
H56	refine vns dense iters 200	200	17	1.442874	200	Cap	8.0
H57	refine vns dense iters 800	800	18	1.396502	329	Local	68.6
H58	refine vns dense all levels	400	18	1.401448	326	Local	72.5
H59	refine vns geometric plus day iters 80	80	17	1.572394	80	Cap	4.9
H60	refine vns geometric plus day iters 200	200	17	1.443957	200	Cap	30.9
H61	refine vns geometric plus day iters 800	800	18	1.399695	346	Local	987.3
H62	refine vns geometric plus day all levels	400	17	1.398578	290	Local	75.1
H63	refine vns wide dense iters 80	80	17	1.567277	80	Cap	2.4
H64	refine vns wide dense iters 200	200	18	1.448894	200	Cap	35.5
H65	refine vns wide dense iters 800	800	18	1.402084	339	Local	135.1
H66	refine vns wide dense all levels	400	18	1.410337	318	Local	157.9
H67	refine rich paired weeks iters 80	80	18	1.562999	80	Cap	4.7
H68	refine rich paired weeks iters 200	200	18	1.421370	200	Cap	27.2
H69	refine rich paired weeks iters 800	800	18	1.403516	261	Local	54.4
H70	refine rich paired weeks all levels	400	18	1.415076	259	Local	81.3
H71	refine vns plus wide iters 80	80	17	1.574522	80	Cap	4.7
H72	refine vns plus wide iters 200	200	17	1.438692	200	Cap	32.3
H73	refine vns plus wide iters 800	800	18	1.404311	291	Local	82.7
H74	refine vns plus wide all levels	400	18	1.413971	261	Local	40.8

Exact shift tuples and selection policy for each historical ID are stored under the matching run name in .scratch/priority-based-selection/hyperparameter-results.json. Local refers only to the tested neighbourhoods and policy, not a global optimum.

Fresh run catalogue

Fresh runs use an 800 cap. Endpoints pass rule replay and two availability recomputations. Start variants are identified by their generation seed. Repeated endpoint hashes confirm determinism. Seconds exclude validation and export work after the search.

ID	Run name	Start	Min	Std	Moves	End	Seconds
F01	default rep1	Loaded	18	1.413971	280	Local	25.1
F02	half steps rep1	Loaded	18	1.391860	319	Local	57.7
F03	wide first rep1	Loaded	18	1.404311	291	Local	44.3
F04	default rep2	Loaded	18	1.413971	280	Local	31.7
F05	half steps rep2	Loaded	18	1.391860	319	Local	61.8
F06	wide first rep2	Loaded	18	1.404311	291	Local	57.3
F07	default rep3	Loaded	18	1.413971	280	Local	49.1
F08	half steps rep3	Loaded	18	1.391860	319	Local	51.4
F09	wide first rep3	Loaded	18	1.404311	291	Local	31.9
F10	default all levels	Loaded	18	1.415233	286	Local	21.2
F11	default first improving	Loaded	18	1.421370	314	Local	14.2
F12	half steps all levels	Loaded	18	1.403198	310	Local	37.9
F13	half steps first improving	Loaded	18	1.408753	391	Local	23.1
F14	dense pairs all levels	Loaded	18	1.401448	326	Local	30.7
F15	dense pairs first improving	Loaded	18	1.404629	344	Local	22.5
F16	ablate half level 1	Loaded	18	1.408595	284	Local	18.9
F17	ablate half level 2	Loaded	18	1.395383	297	Local	23.4
F18	ablate half level 3	Loaded	18	1.396183	303	Local	26.1
F19	ablate half level 4	Loaded	18	1.400332	279	Local	19.9
F20	reverse halves	Loaded	18	1.438537	792	Local	28.7
F21	reverse dense	Loaded	17	1.652663	800	Cap	9.1
F22	halves plus two	Loaded	18	1.391860	319	Local	27.0
F23	halves plus 42	Loaded	18	1.405105	287	Local	25.0
F24	halves plus 21	Loaded	18	1.396183	321	Local	29.3
F25	start 41 default	41	18	1.422469	280	Local	14.6
F26	start 41 half steps	41	18	1.405264	299	Local	22.1
F27	start 41 wide first	41	18	1.405740	294	Local	19.9
F28	start 42 default	42	18	1.415864	305	Local	16.9
F29	start 42 half steps	42	18	1.403357	318	Local	25.8
F30	start 42 wide first	42	18	1.409704	292	Local	20.4
F31	start 43 default	43	18	1.412076	272	Local	13.9
F32	start 43 half steps	43	18	1.400651	307	Local	63.5
F33	start 43 wide first	43	18	1.413182	272	Local	69.0
F34	dense pairs rep1	Loaded	18	1.396502	329	Local	94.9

Source record and parameter definitions

The report derives its numerical claims from local executable scheduling code and run records. Dates refer to the Brisbane study date. SHA256 hashes identify the current source bytes and stored result files.

Production algorithm

savn_priority_based_selection.py

SHA256 0ff76a38ca83a163376564ffaee5cfd7
192ecbdcfd8d5371ecd7cd1b2ff508ea

Source workbook

heavy_maintenance_forecast.xlsx

SHA256 a98a1a3725cf3da075034932aa94e314
d8b99933e2b942b305730a35978667a2

Historical driver

.scratch/priority-based-selection/run_hyperparameter_study.py

SHA256 e6ab3b3e1c05f0c94f0f0fcde3b4c84d
a1cab10a894914c8d729e855bf3e0218

Historical results

.scratch/priority-based-selection/hyperparameter-results.json

SHA256 2cd9c92352da7effeccb8e7e20513140
0e2cbc9ede90a51f26f97f249f7912db

Fresh driver

.scratch/priority-hyperparameter-analysis/run_analysis.py

SHA256 b47ce8603aa9da6fed54cb67e315d4f9
005ff82b25190e78b36f55ea1bab3552

Fresh results

.scratch/priority-hyperparameter-analysis/fresh-results.json

SHA256 53e6ecd271492f6c35b24b95fbeaee0d
fa8994146a57c341454c60fb6b871107

Exact neighbourhood definitions

Configuration	Neighbourhoods in order
Default	-28, +28 | -14, +14 | -7, +7 | -3, +3
Half steps	-28, -14, +14, +28 | -14, -7, +7, +14 | -7, -3, +3, +7 | -3, -1, +1, +3
Dense pairs	-28, +28 | -21, +21 | -14, +14 | -7, +7 | -3, +3 | -1, +1
Wide first	-42, +42 | -28, +28 | -14, +14 | -7, +7 | -3, +3

Vertical bars separate neighbourhoods. Commas separate offsets evaluated within a neighbourhood. Each offset is measured in days. first_level is best improvement in the first improving priority level; all_levels is best improvement across levels; first_improving is the first improving candidate in deterministic order.

Complete neighbourhood catalogue

This catalogue makes the tested search structures explicit without requiring the raw JSON. Duplicate neighbourhood definitions are listed once. Caps and policies are identified in the run tables and selection comparison. Negative offsets move a check earlier; positive offsets move it later within its current legal window.

First run using structure	Offsets in days with levels separated by bars
baseline default	-28,+28 | -14,+14 | -7,+7 | -3,+3
single pm1	-1,+1
single pm2	-2,+2
single pm3	-3,+3
single pm5	-5,+5
single pm7	-7,+7
single pm10	-10,+10
single pm14	-14,+14
single pm21	-21,+21
single pm28	-28,+28
single pm35	-35,+35
single pm42	-42,+42
single pm56	-56,+56
single pm70	-70,+70
vns plus day	-28,+28 | -14,+14 | -7,+7 | -3,+3 | -1,+1
vns plus wide	-42,+42 | -28,+28 | -14,+14 | -7,+7 | -3,+3
vns plus month	-56,+56 | -28,+28 | -14,+14 | -7,+7 | -3,+3
vns early window	-70,+70 | -42,+42 | -28,+28 | -14,+14 | -3,+3
vns weekly	-28,+28 | -14,+14 | -7,+7 | -1,+1
vns fine	-14,+14 | -7,+7 | -3,+3 | -1,+1
vns coarse	-70,+70 | -42,+42 | -28,+28
vns geometric	-32,+32 | -16,+16 | -8,+8 | -4,+4 | -2,+2
vns geometric plus day	-32,+32 | -16,+16 | -8,+8 | -4,+4 | -2,+2 | -1,+1
vns three level	-28,+28 | -7,+7 | -1,+1
vns two level 28 7	-28,+28 | -7,+7
vns two level 14 3	-14,+14 | -3,+3
vns monthish	-30,+30 | -15,+15 | -7,+7 | -3,+3
vns dense	-28,+28 | -21,+21 | -14,+14 | -7,+7 | -3,+3 | -1,+1
vns wide dense	-56,+56 | -42,+42 | -28,+28 | -14,+14 | -7,+7 | -3,+3 | -1,+1
order small first default	-3,+3 | -7,+7 | -14,+14 | -28,+28
order small first weekly	-1,+1 | -7,+7 | -14,+14 | -28,+28
order interleaved	-14,+14 | -28,+28 | -3,+3 | -7,+7
order wide then day	-42,+42 | -1,+1 | -28,+28 | -7,+7
rich paired weeks	-28,-14,+14,+28 | -7,-3,+3,+7
rich one union	-28,-14,-7,-3,+3,+7,+14,+28
rich dense union	-28,-21,-14,-7,-3,-1,+1,+3,+7,+14,+21,+28
rich vns with halves	-28,-14,+14,+28 | -14,-7,+7,+14 | -7,-3,+3,+7 | -3,-1,+1,+3

First run using structure	Offsets in days with levels separated by bars
asymmetric left heavy	-28,-14,+14 | -7,-3,+3
asymmetric right heavy	-14,+14,+28 | -3,+3,+7
ablate half level 1	-28,+28 | -14,-7,+7,+14 | -7,-3,+3,+7 | -3,-1,+1,+3
ablate half level 2	-28,-14,+14,+28 | -14,+14 | -7,-3,+3,+7 | -3,-1,+1,+3
ablate half level 3	-28,-14,+14,+28 | -14,-7,+7,+14 | -7,+7 | -3,-1,+1,+3
ablate half level 4	-28,-14,+14,+28 | -14,-7,+7,+14 | -7,-3,+3,+7 | -3,+3
reverse halves	-3,-1,+1,+3 | -7,-3,+3,+7 | -14,-7,+7,+14 | -28,-14,+14,+28
reverse dense	-1,+1 | -3,+3 | -7,+7 | -14,+14 | -21,+21 | -28,+28
halves plus two	-28,-14,+14,+28 | -14,-7,+7,+14 | -7,-3,+3,+7 | -3,-1,+1,+3 | -2,+2
halves plus 42	-42,+42 | -28,-14,+14,+28 | -14,-7,+7,+14 | -7,-3,+3,+7 | -3,-1,+1,+3
halves plus 21	-28,-21,-14,+14,+21,+28 | -14,-7,+7,+14 | -7,-3,+3,+7 | -3,-1,+1,+3

Structures that share offsets but differ in neighbourhood grouping or order are distinct. Within-neighbourhood offset order is also preserved because first-improving selection depends on it.